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Sex Female | Date of birth 16/07/1993 | Nationality Indian

WORK EXPERIENCE

May 2024-May 2025 **DBT Postdoctoral Research Associate-I**
Microbes and Biology of Infection
Centre for Cellular and Molecular Biology, Hyderabad, India, 500007
Project title: 1. Development of a novel mRNA vaccine against Japanese encephalitis virus
2. Understanding the role of amino acid homeostasis and glycolysis in fungal morphogenesis and its implications in biofilm formation and virulence

EDUCATION AND TRAINING

July 2017 – October 2023 **PhD**
Thesis title: Biofilm modulation and catabolic gene expression in marine bacteria for biodegradation of polycyclic aromatic hydrocarbon (Degree Awarded, October 2023)
Supervisor: Prof Surajit Das
Department of Life Science
National Institute of Technology, Rourkela, Odisha, India 769008
Research area: Biofilm based bioremediation

July 2015 - June 2017 **MSc Biotechnology**
Dissertation project: Studies on genetic polymorphism in Indian germplasms of *Trapa natans* using DNA based molecular marker
Supervisor: Dr Nitish Kumar
Department of Biotechnology,
Central University of South Bihar, Gaya, Bihar, India, 824236
Research area: Plant biotechnology

July 2011 - June 2014 **BSc Biotechnology**
Dissertation project: Callus induction in *Phaseolus vulgaris*, DNA isolation and amplification of ATPase gamma subunit gene
Supervisor: Dr Supriya Shrivastava
Ranchi University, Ranchi, Jharkhand, India, 834001
Subjects studied: Biotechnology, Chemistry, Zoology

TECHNICAL EXPERTISE

- **Cell culture:** Cell lines handled: HEK 293T, RAW 264.7, Cell Transfections, Cell isolation (BMDM), Cell differentiation, Ex vivo assays, Cell line maintenance & cryopreservation, Cell viability & proliferation assays
- **Microbiology:** Bacterial & Fungal culture maintenance and storage, Microbial identification, Microbial genome analysis, Growth curve, Serial dilution, MIC, Cytotoxicity assay, Bioreporter assay, Biofilm assays

- **Molecular Biology:** PCR, qRT-PCR, Gene cloning and manipulation, CRISPR-Cas9 mediated gene editing, Invitro Transcription, RNA modification, LNP formulation, Enzyme Activity, Protein Purification, SDS PAGE, Western Blotting, ELISA
- **Microscopy:** Confocal Laser Scanning Microscopy, Fluorescence Microscopy, Phase Contrast Microscopy, Scanning Electron Microscopy, Atomic Force Microscopy
- **Spectroscopy:** UV-Vis Spectroscopy, Fluorescence Spectroscopy, Attenuated Total Reflection-Fourier Transform Infrared Spectroscopy (ATR-FTIR) spectroscopy, DLS
- **Chromatography:** HPLC, GC-MS, TLC
- **Bioinformatics:** Whole genome analysis, Phylogenetic analysis, Protein structure prediction, Molecular docking, CLUSTALW, Benchling, Snapgene
- **Animals handled:** BALB/c Mice, C57BL6 Mice
- **Other Techniques:** Zeta Potential Analysis, XRD
- **Data Analysis Softwares:** Origin, Graph Pad Prism, Image J, COMSTAT, MEGA, Daime

ACADEMIC ACHIEVEMENTS

Peer-Reviewed Journal Publications

- **Mahto, K. U., & Das, S. (2025).** Exogenous acyl homoserine lactones modulate biofilm formation in *Pseudomonas aeruginosa* PFL-P1 and enhance phenanthrene degradation. *Journal of Chemical Technology & Biotechnology*, 100(10), 2146-2162. <https://doi.org/10.1002/jctb.70023>
- **Mahto, K. U., & Das, S. (2024).** Electroactive biofilm communities in microbial fuel cells for the synergistic treatment of wastewater and bioelectricity generation. *Critical Reviews in Biotechnology*, 45(2), 434-453. <https://doi.org/10.1080/07388551.2024.2372070>
- **Mahto, K. U., Vandana, Priyadarshane, M., Samantaray, D. P., & Das, S. (2022).** Bacterial biofilm and extracellular polymeric substances in the treatment of environmental pollutants: Beyond the protective role in survivability. *Journal of Cleaner Production*, 134759. <https://doi.org/10.1016/j.jclepro.2022.134759>
- **Mahto, K. U., & Das, S. (2022).** Bacterial biofilm and extracellular polymeric substances in the moving bed biofilm reactor for wastewater treatment: A review. *Bioresource Technology*, 345, 126476. <https://doi.org/10.1016/j.biortech.2021.126476>
- **Mahto, K. U., Kumari, S., & Das, S. (2021).** Unraveling the complex regulatory networks in biofilm formation in bacteria and relevance of biofilms in environmental remediation. *Critical Reviews in Biochemistry and Molecular Biology*, 57(3), 305-332. <https://doi.org/10.1080/10409238.2021.2015747>
- **Mahto, K. U., & Das, S. (2021).** Microscopic techniques to evaluate the biofilm formation ability of a marine bacterium *Pseudomonas aeruginosa* PFL-P1 on different substrata. *Microscopy Research and Technique*, 84(10), 2451-2461. <https://doi.org/10.1002/jemt.23799>
- **Mahto, K. U., & Das, S. (2020).** Whole genome characterization and phenanthrene catabolic pathway of a biofilm forming marine bacterium *Pseudomonas aeruginosa* PFL-P1. *Ecotoxicology and Environmental Safety*, 206, 111087. <https://doi.org/10.1016/j.ecoenv.2020.111087>
- **Mahto, K. U., Shaheen, A., Kumari, S., Singh, I. S., & Kumar, N. (2018).** DNA polymorphism analysis of Indian germplasms of *Trapa natans* using RAPD molecular Marker. *Biocatalysis and Agricultural Biotechnology*, 15, 146-149. <https://doi.org/10.1016/j.bcab.2018.06.001>

Book Chapter

- Vandana, Priyadarshane, M., **Mahto, K. U., & Das, S. (2022).** Mechanism of toxicity and adverse health effects of environmental pollutants. In *Microbial Biodegradation and Bioremediation* (pp. 33-53). Elsevier.

Patent

- Das, S., & **Mahto, K. U. (2025).** Development of bacterial biofilm on low-cost substratum for degradation of phenanthrene. The Patent Office, Government of India, Patent No. 567617

Conference Proceedings

- **Mahto, K. U. & S. Das (2018)** Biofilm formation and chemotaxis of marine *Pseudomonas* in biodegradation of polycyclic aromatic hydrocarbons. 59th Annual Conference of Association of Microbiologists of India & International Symposium on Host Pathogen Interaction (09-12 December, 2018), University of Hyderabad, Hyderabad

Seminar/ Workshop Attended

- Workshop on Genomics for Microbial Diversity and Taxonomy- 21st – 25th January 2019 sponsored by MoEFCC India, organised

by Dept. of Life Science, NIT Rourkela, India

- Workshop on Hands on Computational Biology for (Meta) Genomics Analysis- 08th December, 2018 organized by Association of Microbiologists of India (AMI-2018) at University of Hyderabad, Hyderabad, Telangana, India
- Workshop on “Awareness Generation on Intellectual Property Rights” December-27, 2017, organized at National Institute of Technology, Rourkela, Odisha, India in collaboration with the Odisha Department of Science and Technology
- Participated in the “1st International Conference on Human Implications of Biotechnology” 12-14th February, 2016, held at Central University of South Bihar, Patna
- Seminar on “Interdisciplinary Advances in Bio-Technology” held at Ranchi women’s College, Ranchi, Jharkhand, India.
- Bioignition programme on “Exploring the Emerging Paradigms of Biotechnology & Applied Microbiology”- 12-14th February, 2014, held at School of Biotechnology, KIIT University, Bhubaneswar, Odisha

Honours and Awards

- Qualified Graduate Aptitude Test in Engineering (GATE) 2017 in Biotechnology with a score of 414 and All India Rank: 844
- Awarded DBT Research Associateship in 2024. Award Letter No. DBT-RA/2023-2024/Call II/RA/19

REFERENCES

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Research Summary

My PhD research at **National Institute of Technology Rourkela, India**, focused on **elucidating the potential of biofilm-forming marine bacteria for the degradation of polycyclic aromatic hydrocarbons (PAH)**. I isolated several biofilm-forming bacterial strains from PAH-contaminated sediment samples. The biofilm-forming ability of the bacterial strains was assessed qualitatively and quantitatively. Among the several marine bacterial strains isolated, *Pseudomonas aeruginosa* PFL-P1 depicted exceptional biofilm-forming ability. Therefore, the biofilm-forming ability of the strain was analyzed on different substrata such as glass, polystyrene, steel, ceramic, and rubber. Different microscopic techniques were employed such as atomic force microscopy, scanning electron microscopy, and confocal laser scanning microscopy to evaluate the biofilm forming ability (*Microscopy Research and Technique*, 84(10), 2451-2461). Phenanthrene degradation studies carried out using both the planktonic and biofilm modes of growth revealed higher degradation in the biofilm mode. Fluorescence spectroscopic studies confirmed the role of extracellular polymeric substances in the sequestration of pollutants. The emulsification of pollutants in EPS was also confirmed. Thus, it was established that EPS increases the bioavailability of pollutants for increased degradation of pollutants in the biofilm mode of growth (*Journal of Cleaner Production*, 379, 134759). In the natural environment, bacteria occur in polymicrobial communities, and quorum sensing (QS) mediated crosstalk occurs to enhance survivability. Several genera in the Proteobacteria group produce acyl homoserine lactones (AHLs) to orchestrate group behaviors. The interaction between the Las and Rhl QS system with the non-cognate AHLs, C6-HSL, and 3OC8-HSL was studied using molecular docking. Exogenous addition of AHLs improved biofilm formation, motility, cell surface hydrophobicity, and cell auto- aggregation in *Pseudomonas aeruginosa* PFL-P1. The phenanthrene degradation efficiency also increased after the addition of AHLs. Exogenously supplemented AHLs significantly improved the relative expression of QS genes, biofilm matrix gene, and phenanthrene catabolic gene in *P. aeruginosa* PFL-P1 (*Journal of Chemical Technology & Biotechnology*). Further, the whole genome of *Pseudomonas aeruginosa* PFL-P1 was sequenced over the Illumina platform. Gene annotation and pathway analysis revealed several genes involved in xenobiotic biodegradation and metabolism, confirming the utilization of multiple pathways for phenanthrene degradation and potential of the strain for application in diverse bioremediation applications (*Ecotoxicology and Environmental Safety*, 206, 111087). This project provided me with extensive experience in microbial ecology, biofilm biology, and environmental biotechnology, integrating molecular, microscopic, and genomic approaches to elucidate the bioremediation potential of marine bacteria through quorum sensing-regulated polycyclic aromatic hydrocarbon degradation.

Further, during my tenure as a **Postdoctoral Research Associate** at the **CSIR-Centre for Cellular and Molecular Biology (CCMB), Hyderabad, India**, I was involved in two distinct yet complementary research projects that expanded my expertise in RNA therapeutics and microbial pathogenesis.

- **Development of an mRNA-Based Vaccine Against Japanese Encephalitis Virus**
In this multidisciplinary project, I aimed to design, synthesize, and validate a synthetic mRNA vaccine targeting structural proteins of JEV. The antigen sequence was identified from NCBI database and codon-optimized. Further, suitable 5' and 3' UTR sequences were designed and

added to the codon-optimized sequences. The fully designed DNA sequences were synthesized and cloned into pGEM-3zf⁺. Further, mRNA was synthesized using in vitro transcription with modifications (Uridine was replaced with Pseudouridine and 5' mRNA cap analogs were added) to improve stability and translation efficiency. The mRNA constructs were transfected into HEK293T cells, followed by Western blotting to assess protein expression and proper folding of target antigens. Further, murine immunization studies (BALB/c) were carried out to assess the immunogenic potential of the candidate vaccine using ELISA. This project provided me with comprehensive experience in RNA biology, immunology, and translational vaccine research, bridging molecular biology with preclinical application.

- **Understanding the role of amino acid homeostasis and glycolysis in fungal morphogenesis and pathogenesis**

The primary focus of this project was to understand the molecular mechanisms underlying amino acid sensing and regulation in *Candida albicans*, with particular emphasis on their role in fungal morphogenesis and virulence. The study aimed to elucidate how disruptions in amino acid homeostasis and glycolysis influence the morphological plasticity that contributes to pathogenicity. In this project, I employed extensive Differential Interference Contrast (DIC) microscopy to visualize morphogenetic transitions from yeast to pseudohyphal and hyphal cells in nitrogen limiting conditions and glycolysis inhibitory conditions. This project provided me with in-depth experience in fungal molecular biology, nutrient sensing, morphogenetic regulation and virulence in *Candida albicans*.